

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458599.7304 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.022
Transit depth (Rp/Rs)^2	1.93 +/- 0.62 [%]
Orbital Inclination (inc)	81.54 +/- 1.69
Airmass coefficient 1 (a1)	1.284 +/- 0.013
Airmass coefficient 2 (a2)	-0.1358 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	None
Optimal Aperture	4.74
Optimal Annulus	7.91
Transit Duration (day)	0.0467 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.139 ± 0.022 vs 0.1594 (deviation 0.85σ)
Duration fit vs published	0.0467 d vs 0.0512 d (deviation 0.61σ)
Mid-time O-C	-4.2 min
Depth SNR	5.8
Residual scatter	0.99 %

Light curve

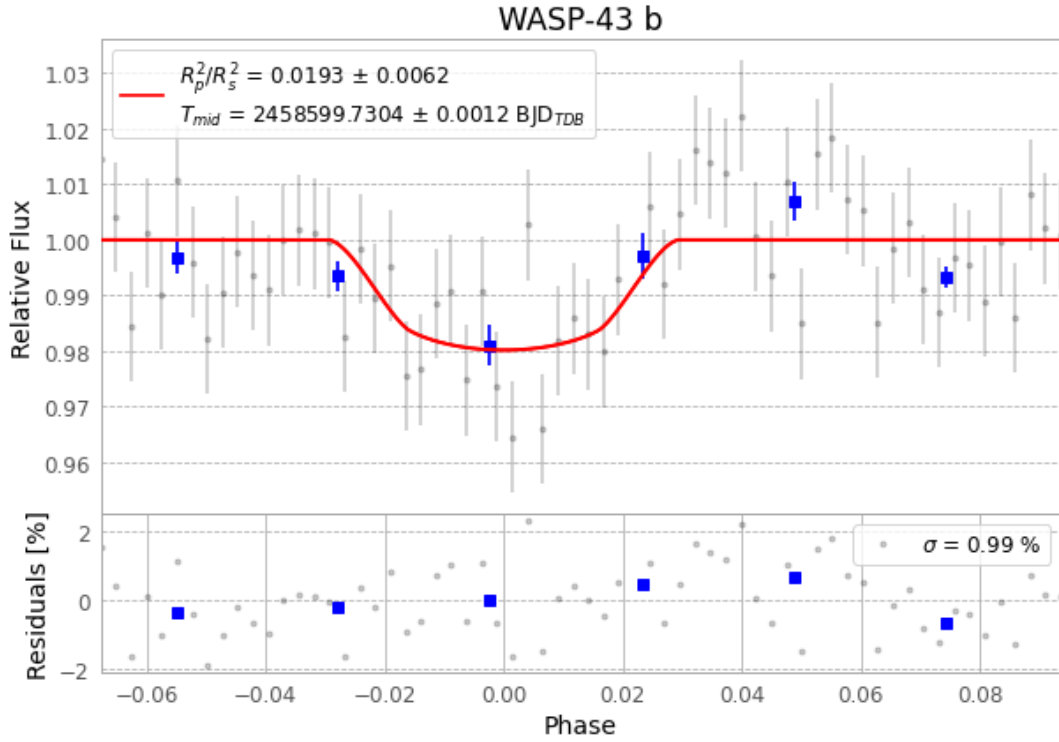


Figure 1 — FinalLightCurve_WASP-43 b_2019-04-25.png (SHA-256 52ef818e16bc2b68...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 733ee9b5c0b440dd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-43190426040615.FITS → WASP-43190426071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190426.log (d048eb095863...), FinalLightCurve_WASP-43 b_2019-04-25.png (52ef818e16bc...), FinalLightCurve_WASP-43 b_2019-04-25.csv (2c58d62a66ba...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:04Z.